FrutaMIDI: capacitive USB MIDI controller

Engineering report

FrutaMIDI engineering project

Repository revision `72f67f0` (interfaces frozen-1)

Abstract

FrutaMIDI is a USB MIDI controller that sends notes when a user touches conductive objects wired to a Raspberry Pi Pico and an Adafruit MPR121 capacitive sensor module, built on a breadboard or on a single-sided home-etched PCB (v1.1). This report collects the requirements, decisions, interfaces, analyses and host tests at revision `72f67f0`. Every number is copied from a cited project file with its label. **No part of the hardware has been built or measured**: analyses, simulations, design-rule checks and host tests are not hardware validation, and the physical validation plan in Section 10 lists what remains to be measured.

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1 Introduction and scope

FrutaMIDI is a USB MIDI controller that turns touches on conductive objects (fruit, plants, water) into MIDI notes. It uses a Raspberry Pi Pico board and an Adafruit MPR121 breakout (product 1982). The project continues an existing repository (v1.0.0 released) and covers two builds: a solderless breadboard and a single-sided PCB for home etching, now at label "FrutaMIDI v1.1" [`management/DECISIONS.md`] (ADR-000, ADR-008).

1.1 Scope

- In scope: electronics design (KiCad, generated by scripts), Pico firmware, the browser sampler (`docs/index.html`), host tests, user documentation and this report [`CLAUDE.md`].
- Out of scope: ordering parts, flashing or powering hardware, and any physical measurement. Hence every [PHYSICAL] requirement is at most PARTIAL or NOT RUN [`management/REQUIREMENTS.md`].
- Hosts: Windows 10, Windows 11 and macOS 12 or later (ADR-007 D-6).

1.2 Document conventions

- Every number carries one label: SOURCE, CALCULATION, SIMULATION, MEASUREMENT or ASSUMPTION, and a citation in brackets to the repository file it is copied from.
- Status is reported only as PASS, PARTIAL, FAIL, BLOCKED or NOT RUN.
- Passing ERC, DRC or a compile is not hardware validation.

2 Requirements and verification status

Source: [`management/REQUIREMENTS.md`], revision draft-5 (T-013), proposed and not yet baselined. Methods: A analysis, S simulation, I inspection, T test. By the project rule, a requirement whose analysis or inspection passes while its physical test is NOT RUN is PARTIAL (ADR-009).

Summary. Of 42 requirements: PASS 17, PARTIAL 18, FAIL 3, NOT RUN 4 [`management/REQUIREMENTS.md`].

- FAIL: REQ-018 VBUS range (the USB low-power-port minimum of 4.40 V is below the Pico's recommended 4.50 V, SOURCE S-01, S-10); REQ-032 core BOM cost, 20.42–30.40 EUR against $\leq$ 30 EUR (CALCULATION, nearly all prices ASSUMPTION); REQ-040 JP1 wrong-position tolerance, kept FAIL with accepted residual risk R-007 (ADR-013) [`requirements/requirements.csv`].
- NOT RUN (physical only): REQ-010 status LED, REQ-011 USB MIDI class compliance, REQ-037 sensing range, REQ-038 false triggers and crosstalk.

The table below is generated from `requirements/requirements.csv` by `reports/report/scripts/csv_to_table.py`; do not edit it by hand.

Table 1: Requirements and verification status (generated from requirements/requirements.csv).

ID	Requirement	Verification	Status	Evidence
REQ-001	**Note On/Off per electrode** Each of the 12 electrode inputs E0–E11 (MPR121 ELE0–ELE11) shall send one MIDI Note On, on the configured channel (default 1) with the configured fixed velocity (default 100, range 1–127), when a touch is detected. It shall send a Note Off for the same note when the touch is released.	I + T [PHYSICAL]	PARTIAL	I: PASS. The chain terminal → series resistor → module electrode is 1:1 (interface_check `pcb.A2.pin2..13`, `pcb.J2.*`). T: NOT RUN.
REQ-002	**Note Off integrity** The Note Off for a pad shall carry the note number sent in its Note On, even if the scale or octave changed in between. All sounding notes shall be released before a recalibration.	T (host unit test) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Hardware test NOT RUN.
REQ-003	**Second sensor (24 pads)** The firmware shall support a second MPR121 at I2C address 0x5B (`NUM_SENSORS = 2`), mapping its electrodes to pads 12–23. This is a breadboard-only option; the PCB carries one module (ADR-007 D-3).	A + T [PHYSICAL]	PARTIAL	A: latency PASS at 400 kHz (12.35 ms $\leq$ 15 ms, jitter 8.35 ms), using the worst-case run-based burst plan (budgets.out §b). T: NOT RUN.
REQ-004	**Polyphonic aftertouch** While a pad is touched, the device shall send Polyphonic Key Pressure (0xAn) with a value 0–127. The value shall map linearly from delta = baseline − filtered, clamped to [12, 120]. It shall be evaluated every 15 ms and sent only when it changes by at least 2.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Aftertouch messages are sent after all Note On/Off of the same loop (ADR-010). Hardware NOT RUN.
REQ-005	**Six scales** The device shall provide 6 scales, selected cyclically: Drum Rack (36–47, octave ignored, the power-on default), major pentatonic, minor pentatonic, major, natural minor and chromatic (root 48). Notes shall be clamped to 0–127.	A + T (host)	PASS	A: notes 24–99 with 12 pads and 24–127 with 24 pads, never clamped (T-009 budgets §d). Host tests PASS (T-040 `test_scales.py`).
REQ-006	**Octave potentiometer** The octave potentiometer shall shift melodic scales by −2..+2 octaves using 5 ADC zones, each V33/5 = 0.660 V wide. The octave shall change only when the reading is more than 0.7 zone from the current zone centre, i.e. 0.2 zone (0.132 V) beyond the border, which is the dead band. The combined ADC noise, DNL, quantisation and supply ripple shall stay below the dead band. A static ADC offset shifts every threshold together and does not consume the dead band. The pot shall have no effect on Drum Rack and shall be enabled by default on the PCB build.	A + T (host) + T [PHYSICAL]	PARTIAL	A: PASS. Error budget 0.0364 V < dead band 0.132 V, margin 0.096 V (budgets A5). Inputs: noise from ENOB 8.7 (SOURCE S-17 Table 627); DNL peak 10 LSB12 (ASSUMPTION; S-17 §4.9.4 and erratum RP2040-E11 give peaks at codes 512/1536/2560/3584, height only in a figure); 10-bit step; ripple 10 mV (ASSUMPTION). The nearest DNL-peak code is 0.049 V from a switch point. The static offset of about 30 mV (S-01 §4.3) is reported, not budgeted (review F-003). Host tests PASS (T-043). Hardware NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-007	**Button short press** A short press of the button (released within 800 ms, sampled every 10 ms) shall select the next scale, and the LED shall blink N times for scale N.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Hardware NOT RUN.
REQ-008	**Button long press recalibration** A press of more than 800 ms shall release all notes and recalibrate all sensors; the LED then blinks 3 times.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Recalibration blocks for 16.8–22.1 ms per sensor at 400 kHz (CALCULATION); this is user-requested and allowed by REQ-016. Hardware NOT RUN.
REQ-009	**Sensor absent or lost** If no MPR121 answers at start, the device shall send no MIDI from it and retry. During play, a sensor shall be marked lost after 3 consecutive failed I2C transactions, or at once when its ECR read-back (every 250 ms) differs from the value read after start-up (a chip that reset itself). On loss the firmware shall release that sensor's notes, show the fast error blink and stop reading it. It shall retry start-up when more than 2 s have passed since the last start attempt (this also covers a second sensor missing at boot). The other sensor shall keep playing.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS: `tests/firmware/logic/test_notes_health.py` (15 tests), within `295 passed` (T-044 handoff). Implementation: `fm_notes.h`/health logic, `startSensor()` = begin → setClock(400000) → configure → ECR read. A self-reset chip releases its notes through the normal touch path before the ECR check, so only one Note Off is sent. Physical NOT RUN (it needs the sensor unplugged or reset during play).
REQ-010	**Status LED** The LED shall give 2 slow blinks at start when a sensor is found, and shall be lit while any pad is touched. On Pico W and Pico 2 W the LED is driven through the CYW43 (WL_GPIO0).	T [PHYSICAL]	NOT RUN	
REQ-011	**USB MIDI class compliance** The device shall enumerate as a USB MIDI 1.0 class device (MIDIStreaming, bulk endpoints). It shall work without vendor drivers on Windows 10, Windows 11 and macOS 12 or later.	T [PHYSICAL]	NOT RUN	
REQ-012	**Device name** The USB product string and the MIDI port name shall be "FrutaMIDI".	I + T [PHYSICAL]	PARTIAL	I: `DEVICE_NAME = "FrutaMIDI"` is passed to `setProductDescriptor` and `setStringDescriptor`. T: NOT RUN.
REQ-013	**MIDI message set** The device shall send only Note On (0x9n, CIN 0x9), Note Off (0x8n, velocity 0, CIN 0x8) and Polyphonic Key Pressure (0xAn, CIN 0xA), on one channel n = MIDI_CHANNEL − 1. Incoming MIDI shall be read and discarded.	I	PASS	`fm_midi.h` constants match S-14 Table 4-1.

ID	Requirement	Verification	Status	Evidence
REQ-014	**Touch-to-MIDI latency** Device latency from the first touch after a pad has been idle for at least 15 ms (DEBOUNCE_MS) to the USB-MIDI packet being available to the host shall be $\leq$ 15 ms worst case, for electrode capacitance $\leq$ 100 pF, with mitigation M1 (one burst read per sensor per aftertouch pass). A re-touch of the same pad within the 15 ms lockout is outside this requirement and is reported separately.	A + T [PHYSICAL]	PARTIAL	A: PASS. 1 sensor 10.72 ms, 2 sensors 12.35 ms at 400 kHz, including the T-044 ECR poll ($\approx$ 0.14 ms per sensor) (CALCULATION, budgets.out §b). Separate term: a re-touch inside the lockout adds up to 15 ms, giving 25.72 ms worst case (review F-011). Design limit C $\leq$ 210 pF. T: NOT RUN (plan item P-2). Flagged for user review (ADR-007).
REQ-015	**Latency jitter** Latency jitter (worst minus best) shall be $\leq$ 10 ms for electrode capacitance $\leq$ 100 pF.	A + T [PHYSICAL]	PARTIAL	A: PASS. 6.72 ms (1 sensor) and 8.35 ms (2 sensors) at 400 kHz
REQ-016	**No blocking during play** During play, no firmware action other than a user-requested recalibration shall block touch processing for more than 10 ms.	I + A	PASS	I: the only `delay()` in firmware is `delay(10)` in `midiBegin()` (setup, FrutaMIDI.ino:111). LED feedback is non-blocking (`fm_blink.h`). A: worst loop 1.59 ms at 400 kHz. Firmware T-041..T-043 is complete (T-044 handoff, 295 host tests passed).
REQ-017	**USB bus power only** The device shall be powered only from USB VBUS through the Pico's micro-USB connector. No other power input shall exist.	I	PASS	Pico pins 39 and 40 are unconnected (`pcb.A1.no_other_nets`).
REQ-018	**VBUS range** The device shall operate with VBUS from 4.40 V to 5.50 V at its connector.	A + T [PHYSICAL]	FAIL	A: VSYS 3.95–5.25 V, inside the 1.8–5.5 V range (S-01 §2.1): PASS. However, the 4.40 V low-power-port minimum (S-10) is below the Pico's recommended VBUS of 5 V $\pm$ 10 % = 4.50 V (S-01 §2.3): budgets check A3b FAIL, so the requirement as stated FAILs (review F-010). Electrically the Pico still runs (VSYS margin 2.15 V). Options: restrict the statement to 4.50–5.50 V, or accept as a documented deviation. T: NOT RUN.
REQ-019	**VBUS current** VBUS current shall be $\leq$ 100 mA (the declared bMaxPower, S-11) for every supported Pico board.	A + T [PHYSICAL]	PARTIAL	A: worst case 55.6 / 70.6 / 60.6 / 75.6 mA for Pico / Pico W / Pico 2 / Pico 2 W. Lower bounds are SOURCE (S-01 Table 3, S-16 Table 2); RP2040 pull-ups 50–80 kΩ are SOURCE S-17. The upper bounds are ASSUMPTIONs. T: NOT RUN (plan item P-3).
REQ-020	**3V3 rail load** The external load on the Pico 3V3(OUT) shall be $\leq$ 300 mA with at least 50 % margin.	A	PASS	10.15 mA worst case with 2 sensors and PCB pull-ups: 96.6 % margin.

ID	Requirement	Verification	Status	Evidence
REQ-021	**MPR121 supply voltage** The MPR121 VDD, fed from the Pico 3V3 through the module LDO, shall be within 2.0–3.6 V.	A	PASS	3.10–3.40 V (CALCULATION; dropout and rail tolerance are ASSUMPTIONs).
REQ-022	**Touchable voltage (SELV)** Every user-touchable conductor shall be at $\leq$ 3.6 V relative to GND and shall be supplied only from the USB SELV source.	A	PASS	Electrodes are charged from MPR121 VDD $\leq$ 3.6 V (S-04), and REQ-017 holds.
REQ-023	**Electrode ESD robustness** Each user-touched electrode terminal shall survive ±4 kV contact discharge without permanent damage. The ±4 kV target is an ASSUMPTION, following IEC 61000-4-2 level 2 (the standard was not read).	I + A + T [PHYSICAL]	PARTIAL	Estimate on ASSUMPTIONs, not a PASS (review F-002). With 3.3 kΩ (R1–R12) plus the IEC 330 Ω source, the peak current at 4 kV is 4 kV / 3.63 kΩ = 1.10 A. That is below the 1.33 A pin limit inferred from the HBM ±2 kV rating, but the 1.33 A limit is itself an ASSUMPTION. Charge and duration are worse than HBM 2 kV: IEC 4 kV through 3.63 kΩ gives τ = 545 ns and Q = 600 nC, against τ = 150 ns and Q = 200 nC for HBM 2 kV (1.5 kΩ, 100 pF), i.e. 3× the charge and 3.6× the time constant (CALCULATION). The MFR-12 3k3 has no datasheet pulse or voltage rating $\geq$ 3.64 kV (ADR-012). 8 kV: risk R-006. Closure only by an ESD-gun test with a pulse-rated resistor (CR-001). T: NOT RUN [PHYSICAL].
REQ-024	**I2C hand-wire fault current** A misplaced SDA or SCL hand wire (including one on a supply hole S1/S2/S6/S7) shall not drive an RP2040 pad beyond its IO current limits.	I + A	PASS	I: 220 Ω series resistors are present between the Pico pads and the bus: pad 31 SDA_PICO → R16 → SDA, pad 32 SCL_PICO → R15 → SCL (interface_check `fw.pcb.PIN_SDA/SCL_vs_pcb`). A: fault current 15.5 mA per line at 3.4 V, 30.9 mA for both lines, $\leq$ IIOVDD_MAX 50 mA (SOURCE S-17 Table 625). Low-level margins: Pico→MPR121 0.138 V, MPR121→Pico 0.235 V (budgets B5/B6, CALCULATION). Statement reworded after ADR-009 D-9: a wire on a supply hole is still physically possible, but it is now current-limited.

ID	Requirement	Verification	Status	Evidence
REQ-025	**Both module revisions** One PCB shall accept both Adafruit 1982 revisions. JP1 bridges ORIG (+3V3 → S1 = VIN) or QT (+3V3 → S7 = VCC). The SDA/SCL wires from J1 go to S5/S6 (original) or S3/S4 (STEMMA QT).	I + T [PHYSICAL]	PARTIAL	I: PASS. JP1 selector S1/+3V3/S7 with silkscreen legend 'ORIG = no white connectors' and 'QT = 2 white connectors' (`pcb.JP1_legend`). J1 1 = SCL, 2 = SDA. hardware/README.md table (JP1, SDA, SCL) = ORIG/S5/S6 and QT/S3/S4, which matches the Eagle-derived S-row (`doc.hw_readme.jp1_and_wires.*`). T: NOT RUN.
REQ-026	**I2C rise time (PCB)** The PCB-build I2C bus shall meet the rise time for its clock: $\leq$ 300 ns at 400 kHz ($\leq$ 1000 ns at 100 kHz), with the module at 0x5A.	A + T [PHYSICAL]	PARTIAL	A: PASS. R13/R14 4.7 kΩ: t_r 156 ns at 60 pF, C_b limit 115.2 pF (systems, with RPU 80 kΩ from S-17); electronics gives 103 ns at 38 pF. The 220 Ω series resistors add 2.2 ns (ASSUMPTION of 10 pF pad capacitance). IO levels with 220 Ω: B5–B8 PASS, minimum margin 0.138 V (S-17 Table 625, S-04 Table 5). T: NOT RUN (plan item P-4).
REQ-027	**Single-sided PCB** The PCB shall be single-sided: copper on B.Cu only, through-hole parts only, no vias, no copper pour, and no wire bridges apart from the 2 I2C configuration wires and the JP1 solder bridge.	I	PASS	pcb_rules_check: all segments on B.Cu; 0 vias, 0 zones, 0 SMD pads.
REQ-028	**Board outline** The board outline shall be 100 × 70 mm.	I	PASS	Outline 100.000 × 70.000 mm. Board label 'FrutaMIDI v1.1' (`pcb.label_v1.1` PASS).
REQ-029	**Home-etch rules** Home-etch rules: signal tracks $\geq$ 1.0 mm ($\geq$ 0.8 mm for short stubs), power tracks $\geq$ 1.2 mm, clearance $\geq$ 0.6 mm (DRC hard minimum 0.5 mm), annular ring $\geq$ 0.4 mm, drills $\geq$ 0.8 mm.	I + A (DRC)	PASS	pcb_rules_check `SUMMARY: 14 checks, 14 PASS, 0 FAIL` on the current PCB (R15/R16 included); minimum annular ring 0.40 mm. Clearance relies on the electronics DRC (`reports/electronics/evidence/T-055/`), which systems did not re-run.
REQ-030	**Screw-terminal map** The PCB shall provide 14 screw terminals at 5.08 mm pitch: GND, E0..E11, GND. Terminal k (k = 2..13) shall reach MPR121 electrode k − 2 through its series resistor.	I	PASS	interface_check J2.T1..T14 and A2.pin2..13.
REQ-031	**Pin map consistency** The firmware PCB pin map (I2C1 SDA GP26 / SCL GP27, button GP22, pot GP28/ADC2) shall match the PCB nets. The breadboard map (I2C0 GP4/GP5, button GP15, pot GP26) shall match README.md.	I	PASS	interface_check: 0 pin mismatches.

ID	Requirement	Verification	Status	Evidence
REQ-032	**Core BOM cost** The core BOM (assembled board, excluding tools, consumables and USB cable) shall cost $\leq$ 30 EUR.	A	FAIL	20.42–30.40 EUR (CALCULATION, now including R15/R16). Nearly all prices are ASSUMPTIONs; the upper bound exceeds the target by 0.40 EUR. Quoted prices from one supplier would close this.
REQ-033	**Firmware build matrix** The firmware shall build without errors for Pico, Pico W, Pico 2 and Pico 2 W, each for the breadboard and PCB variants (FQBN `rp2040:rp2040:<board>:usbstack=tinyusb`, plus `-DFRUTAMIDI_PCB` for PCB), and for `arduino:avr:leonardo` (optional).	T (compile)	PASS	`reports/firmware/T-044/build/*.log`: 9 of 9 logs (Pico, Pico W, Pico 2, Pico 2 W × std/PCB, plus Leonardo) contain 'Sketch uses'. A clean compile is not a hardware validation.
REQ-034	**Browser sampler** The browser sampler `docs/index.html` shall work in Chrome and Edge (Web MIDI) on Windows and macOS. It shall map its 12 pads to notes 36–47 by default, treat Note On with velocity 0 as Note Off, send unmapped notes to the synthesiser, apply aftertouch (0xA0) to the filter, and offer a keyboard fallback and dropped-file samples.	T	PARTIAL	Automated browser tests with a stubbed Web MIDI PASS (T-020, T-021 handoffs). A real browser, a real device and audible output are NOT RUN.
REQ-035	**English language** All project files, code, comments, reports, generated PCB text, fabrication file names and user-visible strings shall be in English.	I	PASS	I: 0 Portuguese-word matches in firmware (`.ino`, `fm_*.h`), gen_*.py, docs/index.html, README.md, hardware/README.md and docs/manual/MANUAL.md. Silkscreen is English (SCALE, OCTAVE, ORIG, QT, legend, 'copper: bottom side'); fabrication file names are English; no broken relative links (`doc.links.*` PASS).
REQ-036	**User and assembly manual** The user and assembly manual shall match the current PCB. It shall cover both module revisions, the JP1 selector and data wires, flashing, first start, a valid pre-power continuity checklist and troubleshooting.	I	PASS	I: hardware/README.md (assembly) matches the current PCB: JP1 and two wires, R15/R16 220 Ω, JP1 legend, a continuity test that accounts for the 3.3 kΩ resistors, English silkscreen names, valid links (interface_check documentation `17 checks, 17 PASS, 0 FAIL`). README.md documents the breadboard 4.7 kΩ pull-ups. docs/manual/MANUAL.md (user manual) refers to JP1 and covers BOOTSEL/UF2 and troubleshooting. The inspection covered consistency with the design, not readability.

ID	Requirement	Verification	Status	Evidence
REQ-037	**Sensing range (physical)** [PHYSICAL] Touch detection shall work on objects from a bare pad (about 20 pF) to large fruit or a glass of water (up to about 500 pF, ASSUMPTION), with leads of 20–50 cm, after recalibration.	T [PHYSICAL]	NOT RUN	A: the MPR121 can charge up to about 1203 pF (CALCULATION). Latency targets hold up to 210 pF at 400 kHz. AUTOCONFIG limits USL/TL/LSL = 200/180/130 at design VDD 3.2 V and SCTS = 0 (0x7C D7) are now SOURCE (ADR-010/011, MPR121 datasheet S-006 in management/SOURCES.md). T: NOT RUN.
REQ-038	**False triggers and crosstalk (physical)** [PHYSICAL] There shall be no false Note On in 10 min idle with all 12 electrodes connected to objects, and no crosstalk when touching a neighbouring electrode.	T [PHYSICAL]	NOT RUN	
REQ-039	**JP1 cannot short 3V3** No position of the JP1 3V3 selector (ORIG, QT, open) shall connect +3V3 to GND.	I	PASS	S1 and S7 are never GND in either revision (`pcb.JP1_no_gnd_reachable`).
REQ-040	**JP1 wrong-position tolerance** A wrong JP1 position (ORIG on a STEMMA QT module, or QT on an original module), or both sides bridged, shall not apply a voltage outside the MPR121 absolute maximum ratings (inputs $\leq$ VDD + 0.3 V, S-04 Table 3), and shall not short +3V3 through a module output.	A	FAIL	A: still FAIL as stated. The ORIG bridge on a QT module puts +3V3 on ADDR, and the QT bridge on an original module puts it on IRQ, both above VDD + 0.3 V on an unpowered module. Both sides bridged on an original module ties +3V3 to the IRQ open drain. Mitigation per ADR-009 D-10: silkscreen legend present (`pcb.JP1_legend` PASS), manual warning, and the firmware 'sensor not found' LED pattern. Residual risk R-007 accepted by the manager.
REQ-041	**I2C rise time (breadboard, with pull-ups)** The breadboard-build I2C bus, with the two 4.7 kΩ pull-ups required by ADR-009 D-11, shall meet the rise time for 400 kHz.	A + T [PHYSICAL]	PARTIAL	A: PASS. t_r 156 ns at 60 pF with 4.7 kΩ (CALCULATION). Without them, 452 ns: FAIL above 39.8 pF. I: README.md lists the two 4.7 kΩ pull-ups in its BOM and wiring table. T: NOT RUN (plan item P-4).
REQ-042	**Shared-note reference counting** When two or more pads map to the same MIDI note, the note shall be sent once (Note On on the first pad), stay sounding while any of those pads is held, and get its Note Off only when the last of them is released (reference counting). Aftertouch for that note shall be the maximum of the holding pads' values, with a final pressure of 0 on the last release.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS: `tests/firmware/logic/test_notes_health.py` (reviewer sequence, `releaseAll`), `firmware/FrutaMIDI/fm_notes.h` (T-044, 295 passed). Physical NOT RUN.

Status counts (42 requirements): PASS 17, PARTIAL 18, FAIL 3, NOT RUN 4.

3 Architecture and decisions

3.1 System overview

Touch path: electrode object → lead → screw terminal → 3.3 kΩ series resistor (PCB) → MPR121 electrode pin; the MPR121 reports touch bits and filtered/baseline data over I^2C at 400 kHz to the Pico, which sends USB MIDI 1.0 messages to the host (browser sampler or DAW) [`management/INTERFACES.json`] [`reports/systems/architecture_options.md`].

3.2 Architecture options

Three options were compared on the same criteria [`reports/systems/architecture_options.md`]; option A was chosen (ADR-006).

Criterion	A: Pico + MPR121 (chosen)	B: Pico native sensing	C: Pro Micro + MPR121
Core BOM cost	23.60–34.60 EUR (CALCULATION, budgets.out, revision 1)	14.20–22.20 EUR (CALCULATION from ASSUMPTION prices)	22.60–34.60 EUR (ASSUMPTION clone price)
Sensing with fruit/water	Good; hardware autoconfiguration and baseline filter	Unproven; filtering in firmware	Same as A
Latency floor	4 ms at the sensor	Potentially below 1 ms (ASSUMPTION)	Same as A
Constraint compliance	Uses the fixed bought-in parts	Drops the Adafruit 1982 (definition change)	Drops the Pico family (definition change)
Risk level	Low–medium	High	Medium

The BOM figure for A above is the T-008 value; the current core BOM is 20.42–30.40 EUR (CALCULATION, REQ-032, Section 2).

3.3 Decisions

Summary of [`management/DECISIONS.md`] (ADR-000 to ADR-017). Decisions taken by the manager under ADR-001 are flagged there for later user review.

ADR	Decision	Basis / evidence
ADR-000	Project folder, all files in English, existing paths kept, commit rules.	User, Phase A
ADR-001	Run gates G1–G4 end to end; publishing a release still needs explicit user approval.	User
ADR-002	Firmware is the source of truth for the long press: docs state 0.8 s (firmware `> 800` ms).	T-030 handoff
ADR-003	No “safe to touch” claim without evidence; wording follows the electronics analysis.	T-050
ADR-004	English file and folder names (`hardware/fabrication/`, `images/`).	T-051
ADR-005	Sampler: learning a used note swaps the two pads; notes flash before audio is enabled; omni channel; English UI and silkscreen.	T-020
ADR-006	Architecture option A (Pico + Adafruit MPR121 breakout); MPR121 sets a ≥ 4 ms sensing floor.	architecture_options.md

ADR	Decision	Basis / evidence
ADR-007	Latency $\leq$ 15 ms worst case and jitter $\leq$ 10 ms (relaxed, flagged); burst read M1; 24 pads breadboard only; JP1 3-pad selector replaces the 3V3 wire; hosts; BOM target $\leq$ 30 EUR; English.	T-008
ADR-008	3.3 kΩ series resistor per electrode (recorded as PASS ±4 kV, later downgraded to an estimate by review F-002; PARTIAL 8 kV, risk R-006); keep 400 kHz with 4.7 kΩ pull-ups; label v1.1.	analysis.md
ADR-009	PARTIAL policy; 220 Ω I^2C series resistors (fault current $\leq$ 15 mA); JP1 silkscreen and warnings (R-007); breadboard 4.7 kΩ pull-ups.	T-010
ADR-010	Run-based burst reads (12 pads 945 µs vs 2430 µs, CALCULATION); AUTOCONFIG limits 200/180/130 at design VDD 3.2 V; SCTS = 0.	T-041
ADR-011	MPR121 register facts confirmed from the datasheet (S-006); three firmware ASSUMPTIONs become SOURCE.	T-042, manager
ADR-012	No datasheet-rated pulse resistor found; MFR-12 3k3 kept, REQ-023 PARTIAL, candidate CR-001; R15/R16 220 Ω accepted.	T-054, T-055
ADR-013	Interfaces frozen as frozen-1; REQ-040 stays FAIL with accepted risk R-007.	Manager
ADR-014	All 16 review findings of 72f67f0 dispositioned (13 FIX, 1 ACCEPTED, 2 info FIX); fixes made on main before the baseline; no frozen interface changed.	Review T-060
ADR-015	BOM part numbers: J2 Phoenix Contact 1715721 (×7); RV1 named by series (RK09K, order code not sourced, PARTIAL accepted); female headers generic; SW1 Omron B3F-1000.	T-056
ADR-016	REQ-018 stays FAIL with a documented deviation (low-power USB port 4.40 V below the Pico's recommended 4.50 V; guide recommends a standard port or powered hub); documentation-only interface change applied as frozen-2.	T-012
ADR-017	Firmware review fixes: per-note reference counting (F-004), sensor health with loss detection and retry (F-005), `setClock` after every sensor `begin()` (F-006).	T-044

Change candidate. CR-001 (not opened): replace the MFR-12 3k3 by a pulse-rated resistor, for example Vishay MBB 0207, if its datasheet confirms $\geq$ 3.64 kV at 3.3 kΩ; the 4 kV case puts about 3.6 kV across the resistor (CALCULATION) [`changes/CR-001-candidate.md`] [`sim/electronics/calc/calc_pulse.out`].

4 Interfaces

Source: [`management/INTERFACES.json`], revision `frozen-2` (frozen-1 of 2026-11-03, ADR-013, plus a documentation-only change request from T-012, ADR-016; frozen on 2026-11-04). Changes go through the engineering change process.

Interface	Summary	Requirements
IF-USB-MIDI	Full-speed USB, Audio/MIDIStreaming USB MIDI 1.0, bulk IN/OUT 64-byte packets; bus powered, bMaxPower 100 mA (SOURCE S-11); product string "FrutaMIDI". Sends Note On (0x9n), Note Off (0x8n) and Polyphonic Key Pressure on channel `MIDI_CHANNEL` (default 1, range 1–16); incoming MIDI is read and discarded. Hosts: Windows 10/11, macOS 12 or later.	REQ-011, 012, 013, 019

Interface	Summary	Requirements
IF-I2C	400 kHz; sensor 0 at 0x5A, optional sensor 1 at 0x5B (breadboard). PCB pull-ups R13 (SCL), R14 (SDA) 4.7 kΩ to +3V3, plus module 10 kΩ pull-ups and BSS138 shifters. 3.3 V logic. Series R15 (SCL) and R16 (SDA) 220 Ω on the Pico side; fault current 15.5 mA per line (CALCULATION at 3.4 V, ≤ 50 mA IIOVDD, SOURCE S-17).	REQ-021, 025, 026, 031
IF-PICO-PINS	Breadboard: SDA GP4 (pin 6), SCL GP5 (pin 7), button GP15 (pin 20), pot GP26/ADC0 (pin 31, optional), 3V3 pin 36. PCB: SDA_PICO GP26 (pin 31) via R16, SCL_PICO GP27 (pin 32) via R15, button GP22 (pin 29), pot GP28. AGND (pin 33) is the analogue ground return (SOURCE S-01 §2.1). LED GP25 (Pico, Pico 2) or CYW43 WL_GPIO0 (W boards) (SOURCE S-02).	REQ-017, 031
IF-TERMINALS	J2: 14 terminals at 5.08 mm: GND, E0–E11, GND. Terminal k ($k = 2..13$) reaches MPR121 ELE($k-2$) through R1–R12 3.3 kΩ; protection at ±4 kV is an estimate on ASSUMPTIONs, not a pass (review F-002); PARTIAL at 8 kV (R-006).	REQ-022, 023, 030
IF-MPR121-MODULE	Socket A2: 1×13 electrode row and 1×7 power/I²C row S1–S7. JP1: pad 1 S1 (ORIG), pad 2 +3V3, pad 3 S7 (QT); a wrong choice puts +3V3 on ADDR or IRQ, never GND (REQ-039 PASS, REQ-040 FAIL). J1: pad 1 SCL, pad 2 SDA; wires to S5/S6 (original) or S3/S4 (STEMMA QT).	REQ-024, 025
IF-CONTROLS	SW1 to GND with internal pull-up, sampled every 10 ms; long press > 800 ms recalibrates, short press selects the next scale. RV1 10 kΩ linear: 10-bit ADC read every 30 ms, 5 zones of 0.66 V, hysteresis 0.7 zone, dead band 0.132 V against an error budget of 0.0364 V (CALCULATION, budgets.md).	REQ-006, 007, 008
IF-FW-BUILD	arduino-cli 1.5.1, arduino-pico 6.3.1, Adafruit MPR121 1.2.1, BusIO 1.17.4, MIDI Library 5.0.2. FQBN `rp2040:rp2040:<board>:usbstack=tinyusb` for Pico, Pico W, Pico 2, Pico 2 W; `-DFRUTAMIDI_PCB` selects the PCB pin map; optional `arduino:avr:leonardo`. Artifacts `FrutaMIDI-<board>.uf2` and `FrutaMIDI-pcb-<board>.uf2`.	REQ-033

Board: outline 100 × 70 mm, copper on B.Cu only, label "FrutaMIDI v1.1", fabrication folder `hardware/fabrication`.

5 Electronics analysis

Sources: [`reports/electronics/analysis.md`] (T-050/T-052), handoffs T-053, T-054 and T-055 in [`management/handoffs/electronics-engineer/`], raw outputs in [`sim/electronics/calc/`]. Nothing in this section is a measurement.

5.1 Schematic and PCB checks

The schematic and PCB are generated by `hardware/scripts/` and checked by `build_all.sh` with kicad-cli 10.0.4. After T-055 [`reports/electronics/evidence/T-055/`]:

- ERC: "Found 0 violations".
- DRC with schematic parity: "Found 0 violations", "Found 0 unconnected items", "Found 0 schematic parity issues".
- Home-etch rules (`sim/systems/pcb_rules_check.py`): "SUMMARY: 14 checks, 14 PASS, 0 FAIL".

- Interface check at `72f67f0`: "SUMMARY: 78 checks, 78 PASS, 0 FAIL" (hardware/firmware 61/61, documentation 17/17) [`sim/systems/interface_check.out`]. The interim T-055 result (2 expected FAILs on the Pico SDA/SCL nets behind R16/R15) was resolved by the systems update in T-011.

These are design-rule results, not hardware validation.

5.2 Datasheet checks and budgets

- Pico pins (GP26/GP27 I^2C1 on pins 31/32, GP22 pin 29, GP28 ADC2 pin 34) match the firmware PCB map: PASS (SOURCE EE-S08).
- MPR121 VDD through the module regulator: AP2112 ≈3.295 V typical, ≥3.19 V worst; MIC5225 ≈3.25 V typical, ≥3.10 V worst; inside 2.0–3.6 V: PASS (CALCULATION).
- 3V3 load: 4.45 mA worst case, board power 14.7 mW (CALCULATION, electronics). Systems gives 10.15 mA worst case with 2 sensors and PCB pull-ups, 96.6 % margin to 300 mA (CALCULATION) [`reports/systems/budgets.md`] [`management/REQUIREMENTS.md`] (REQ-020).
- VBUS current worst case 55.6 / 70.6 / 60.6 / 75.6 mA for Pico / Pico W / Pico 2 / Pico 2 W against 100 mA (CALCULATION; upper bounds ASSUMPTION) [`reports/systems/budgets.md`].

5.3 Bill of materials

Part numbers from [`hardware/fabrication/bom-kicad.csv`] (ADR-015): A1 Raspberry Pi SC0915; A2 Adafruit 1982; R1–R12 Yageo MFR-12FTF52-3K3; R13/R14 MFR-12FTF52-4K7; R15/R16 MFR-12FTF52-220R; J2 Phoenix Contact 1715721 (×7); RV1 Alps Alpine RK09K series, 10 kΩ linear, vertical (order code not sourced); SW1 Omron B3F-1000; female headers J4–J7 generic. Core BOM cost 20.42–30.40 EUR against ≤ 30 EUR: FAIL (REQ-032, CALCULATION on mostly ASSUMPTION prices) [`reports/systems/budgets.md`].

5.4 Electrode path and ESD (critical block)

Model `sim_electrode.py` (SIMULATION): constant-current charge into a 17 pF pin plus series R plus external C, swept over R = 0–100 kΩ and C = 10–2000 pF; the discharge model is an ASSUMPTION [`reports/electronics/analysis.md`].

Variant	Peak 4 kV	Peak 8 kV	Sensitivity	Verdict
a) none	15.0 A	30.0 A	1.000	FAIL (11× over)
b) 3.3 kΩ series (selected)	1.10 A	2.20 A	0.904	Estimate at 4 kV (review F-002), PARTIAL 8 kV
b') 10 kΩ series	0.39 A	0.77 A	0.679	FAIL (sensitivity)
c) 3.3 kΩ + TPD4E05U06	1.10 A	2.20 A	≈0.90	FAIL (SMD constraint)
d) 3.3 kΩ + 2×1N4148	1.10 A	2.20 A	≈0.90	PARTIAL (12 jumpers)

- ESD model: IEC 150 pF/330 Ω; pin limit 1.33 A inferred from HBM 2 kV (ASSUMPTION). Minimum R: 2.67 kΩ at 4 kV, 5.67 kΩ at 8 kV. Design rule 2.7–3.3 kΩ; chosen 3.3 kΩ (Yageo MFR-12FTF52-3K3).
- With 3.3 kΩ the smallest detectable touch is 0.5 pF on 10 pF and 19.9 pF on 1000 pF (19.8 pF without R) (SIMULATION).
- Resistor pulse rating (T-053/T-054): the peak across R is 3636 V at 4 kV and 7273 V at 8 kV (CALCULATION) [`sim/electronics/calc/calc_pulse.out`]. The MFR-12 is rated 400 V maximum overload with no microsecond pulse rating: FAIL against the datasheet.

No candidate with a datasheet-confirmed ≥3.64 kV rating was found (ADR-012, CR-001). REQ-023 stays PARTIAL.

- The breadboard build has no electrode protection beyond the MPR121's internal structures (HBM ±2 kV, SOURCE EE-S01), because the module wires ELE0–11 straight to its header (T-055).

5.5 I²C bus

- At e4668d4 the rise time with only the module 10 kΩ pull-ups was 322 ns at 38 pF against 300 ns: FAIL (CALCULATION, bus C partly ASSUMPTION). With R13/R14 4.7 kΩ: 103 ns at 38 pF (electronics) and 156 ns at 60 pF (systems, RPU 80 kΩ SOURCE S-17): PASS (CALCULATION) [`reports/systems/budgets.md`].
- Series R15/R16 220 Ω: fault current 15.5 mA per line, 30.9 mA for both, against 50 mA (CALCULATION). Low-level margins (authoritative, RP2040 datasheet levels SOURCE S-17): 0.138 V when the Pico drives low (VIL 0.93 V at the MPR121) and 0.235 V when the MPR121 drives low (VIL 0.8 V at the Pico): PASS (CALCULATION) [`reports/systems/budgets.md`]. The earlier electronics figures (0.218 V and 0.286 V, with RP2040 levels as ASSUMPTION, T-055) are superseded interim values.
- Breadboard without the 4.7 kΩ pull-ups: 452 ns at 60 pF, FAIL above 39.8 pF; hence ADR-009 D-11 (CALCULATION).

5.6 Safety statement

Each PCB electrode has a 3.3 kΩ series resistor. Its protection against IEC 61000-4-2 contact discharge at ±4 kV is an estimate resting on ASSUMPTIONs (the pin limit is inferred from HBM 2 kV), not a pass; it compares peak current only, and the resistor itself is not pulse-rated for the resulting voltage (review finding F-002 [`reports/review/72f67f0/findings.md`]). Only an ESD-gun test, with a pulse-rated resistor (CR-001), can close it. The design is not evaluated beyond that, and not for touching while connected to mains-powered equipment (ADR-003, ADR-012).

6 Firmware design and tests

Sources: [`reports/firmware/coverage.md`] and the T-040 to T-043 handoffs in [`management/handoffs/firmware-engineer/`].

6.1 Design

- One sketch `firmware/FrutaMIDI/FrutaMIDI.ino` with host-testable header-only logic `fm_*.h`: scales (`fm_scales.h`, English names), button (`fm_button.h`), debounce, aftertouch, potentiometer, non-blocking LED blinker (`fm_blink.h`), MPR121 burst reads (`fm_mpr121.h`).
- Pads: Note On at velocity 100 on channel 1 by default; Note Off always carries the stored note; debounce is a 15 ms lockout per pad.
- Aftertouch: Polyphonic Key Pressure 0–127 from delta 12–120; one 0 is sent before Note Off and on a fall to the floor (T-041).
- Scales: 6, Drum Rack (36–47, octave ignored) default; octave $-2..+2$ by potentiometer with 0.7-zone hysteresis.
- Shared notes (T-044, ADR-017): a reference count per note; Note On when the first pad takes a note, Note Off only when the last pad releases it, aftertouch is the maximum of the holders.

- Sensor health (T-044): 3 consecutive failed transactions, or a changed ECR read-back (every 250 ms), mark a sensor lost; its pads are released, the LED shows the fast error blink, and the sensor is restarted after more than 2000 ms (SOURCE `fm_sensor_health.h`). `setClock(400000)` now follows every sensor `begin()` (F-006).
- Defect found (ADR-017): on an I^2C NACK the Adafruit library's `touched()` returns 0x0FFF, i.e. all 12 pads touched, so a lost sensor would have sent 12 Note Ons. The sketch now reads the status registers itself and checks the transaction result [`management/handoffs/firmware-engineer/T-044.md`].
- Button: long press > 800 ms recalibrates (releases all notes, restarts sensors, 3 blinks); short press selects the next scale.
- LED patterns are non-blocking: pads stay live while the LED blinks (T-041). `startSensors()` still blocks the loop during a user-requested recalibration (kept by decision; duration to be measured).
- MPR121 configuration: AUTOCONFIG with USL/TL/LSL = 200/180/130 at design VDD 3.2 V and SCTS = 0 (ADR-010, ADR-011, SOURCE MPR121 datasheet S-006); I^2C stays at 400 kHz; run-based burst reads (worst mask 42 bus bytes in 8 transactions, CALCULATION) [`reports/systems/budgets.md`].

6.2 Latency budget

Budgets revision 5 (T-013, includes the ECR health poll) [`reports/systems/budgets.md`] [`sim/systems/budgets.out`], all CALCULATION for $C \leq 100$ pF at 400 kHz: worst case 10.72 ms (1 sensor) and 12.35 ms (2 sensors) against ≤ 15 ms; jitter 6.72 ms and 8.35 ms against ≤ 10 ms: PASS. A re-touch of the same pad within the 15 ms lockout adds up to 15 ms, giving up to 25.72 ms (1 sensor); REQ-014 is stated for a pad idle for at least 15 ms. At 100 kHz the jitter (1 sensor) and both 2-sensor checks FAIL, so the bus stays at 400 kHz (ADR-008).

6.3 Build and tests

Task	Result line (MEASUREMENT)	Status	Evidence
T-040	Host tests "213 passed in 1.33s"; 14/14 mutants fail the suite; compile matrix 9/9	PASS	`reports/firmware/T-040/`
T-041	Host tests "267 passed in 1.79s"; compile matrix 9/9	PASS	`reports/firmware/T-041/`
T-042	Host tests "268 passed in 1.61s"; datasheet check BLOCKED, later closed by ADR-011	PARTIAL	`reports/firmware/T-042/`
T-043	Host tests "268 passed in 1.97s"; rpipico PCB "Sketch uses 90700 bytes (4%) of program storage space", "Global variables use 17632 bytes (6%)"	PASS	`reports/firmware/T-043/checks.log`
T-044	Host tests "295 passed in 1.40s"; mutation check 10/10 killed; 9/9 builds exit 0 (rpipico PCB "Sketch uses 91196 bytes (4%)")	PASS	`reports/firmware/T-044/`

A clean compile and host tests are not hardware validation. The host loop harness copies the call order of `loop()` and does not run the sketch itself; the timing granularity on target depends on the loop period, which has not been measured [`reports/firmware/coverage.md`].

7 Host software and tests

Source: `docs/index.html` (browser sampler, GitHub Pages), handoffs T-020 and T-021 in [`management/handoffs/software-engineer/`], raw outputs in [`reports/software/`].

7.1 Design

- Web MIDI input, omni channel; 12 pads mapped to notes 36–47 by default; Note On with velocity 0 is treated as Note Off (REQ-034).
- Learning a note already used by another pad swaps the two pads' notes; notes received before audio is enabled flash the pad and emphasise the enable-sound button (ADR-005). All UI strings are in English (`lang="en"`).

7.2 Tests

- T-020: `run_pytest tests/host` with Python 3.9.6, pytest 8.4.2, node v26.4.0, playwright-core 1.64.1, Google Chrome 156.0.8078.74: "37 passed in 69.20s" (MEASUREMENT) [`reports/software/T-020/pytest_tests_host_verbose.txt`].
- T-021: same tools, Chrome headless: "102 passed in 56.45s" (MEASUREMENT), including the ADR-005 behaviour tests, a rendered language test over 8 states and mutation checks [`reports/software/T-021/pytest_tests_host_verbose.txt`].

The tests use a stubbed Web MIDI API in headless Chrome. No real MIDI device, no audible output and no person looking at the page in a real browser have been tested (T-020, T-021).

8 Independent review findings

Review of `72f67f0` (task T-060): 16 findings (0 critical, 3 major, 9 minor, 4 info). Dispositions are in [`reports/review/72f67f0/findings.md`]; ADR-014 records 13 FIX, 1 ACCEPTED and 2 info FIX. Re-review pending.

TO BE FILLED FROM EVIDENCE. Per-finding table (ID, severity, disposition, evidence) and the re-review result.

9 Limitations

No hardware tests. Nothing has been built, flashed, powered or measured. Building and measuring hardware is out of scope for this project (risk R-004, "Certain", "High") [`management/RISKS.md`]. Every [PHYSICAL] requirement is therefore PARTIAL or NOT RUN, and no statement in this report means that the hardware works.

Analysis-only results. The following results rest only on calculation, simulation, design-rule checks or host tests: latency and jitter (REQ-014/015); VBUS range and current (REQ-018/019); I^2C rise time and logic levels (REQ-026/041, REQ-024); electrode ESD (REQ-023); sensitivity with the series resistors (SIMULATION); ERC/DRC/parity and home-etch rules; firmware behaviour (host tests on a copy of the loop order); sampler behaviour (headless Chrome with a stubbed Web MIDI API).

Open FAILs and accepted risks. REQ-018 (4.40 V low-power-port VBUS minimum below the Pico's recommended 4.50 V; documented deviation, ADR-016); REQ-032 (BOM upper bound 30.40 EUR > 30 EUR, prices mostly ASSUMPTION); REQ-040 (wrong JP1 position exceeds MPR121 input ratings on an unpowered module; accepted risk R-007); R-006 (8 kV contact exceeds the inferred pin limit by 1.65×, accepted); the electrode resistor is not datasheet-rated for the ESD pulse voltage (ADR-012).

Main assumptions (each to be retired by the plan in Section 10):

- Component prices (BOM, REQ-032) [`reports/systems/budgets.md`].
- Electrode capacitance: objects up to about 500 pF (REQ-037); 1 m lead 30–50 pF; hand 100–300 pF [`reports/electronics/analysis.md`].
- ESD: ±4 kV target following IEC 61000-4-2 level 2 (the standard was not read); MPR121 pin limit 1.33 A inferred from HBM 2 kV; electrode discharge model; flashover at about 3 kV/mm; a 1.2/50 µs rating taken as conservative for the ≈0.5 µs pulse.
- Board currents (upper bounds), Schottky diode drop and regulator efficiency [`reports/systems/budgets.md`].
- I^2C bus capacitance; Pico pin capacitance; in the electronics calculation, RP2040 VIL 0.8 V and VOL 0.4 V and BSS138 $R_{DS(on)} \leq 10\,\Omega$ (T-055).
- MIC5225 typical dropout from a search extract only [`reports/electronics/analysis.md`].
- Firmware timing: loop period "well under 15 ms", button bounce < 10 ms, header and pot pin sizes for the 1.0 mm holes [`reports/firmware/coverage.md`].

Process limitations. Requirements are draft-5 (T-013), not baselined. The independent review is still running (Section 8). The interface check reports two expected FAILs that systems must update (T-055). Some user-documentation statements (prices, OS behaviour, etching process values) are not traced to a source (documentation handoffs T-030 to T-035).

10 Physical validation plan

No part of the hardware has been measured. Each row is a physical test still to run. Pass criteria are the requirement thresholds in [`management/REQUIREMENTS.md`]; where a requirement gives no threshold, the analysis prediction is the criterion and is labelled. Items P-1 to P-6 are the systems plan [`management/handoffs/systems-engineer/T-011.md`]; hardware items from firmware are in [`reports/firmware/coverage.md`]; electronics next tests are in the T-050 to T-055 handoffs. All rows are NOT RUN.

Table 7: Physical validation plan (all NOT RUN).

ID	What to measure	How / instrument	Pass criterion	REQ
V-01 (P-1)	Object plus lead capacitance	LCR meter on typical objects (bare pad, fruit, glass of water) with 20–50 cm leads	Touch detected after recalibration from about 20 pF to about 500 pF (ASSUMPTION range in REQ-037)	REQ-037
V-02 (P-2)	Touch-to-USB latency and jitter	Logic analyser on a test electrode, SDA/SCL and USB	Worst case ≤ 15 ms; jitter ≤ 10 ms, for $C \leq 100$ pF	REQ-014, 015
V-03 (P-3)	VBUS current per Pico variant	USB inline current meter; Pico, Pico W, Pico 2, Pico 2 W	≤ 100 mA	REQ-019
V-04	VBUS range	Adjustable bench supply at 4.40 V and 5.50 V at the connector	Device enumerates and plays at both ends	REQ-018
V-05 (P-4)	I^2C rise time and low levels, PCB and breadboard	Oscilloscope on SDA/SCL at the Pico and at the module	Rise ≤ 300 ns at 400 kHz; low levels below VIL (0.93 V at the MPR121, 0.8 V at the Pico)	REQ-026, 041
V-06	Misplaced data wire current	Wire J1 SDA/SCL to a supply hole; current meter in series	≤ 15.5 mA per line (CALCULATION); within RP2040 IO limits	REQ-024
V-07 (P-5)	Host enumeration	Windows 10, Windows 11, macOS 12 or later; MIDI monitor	No vendor driver; name "FrutaMIDI"; Note On/Off and poly aftertouch on channel 1 only	REQ-011, 012, 013

ID	What to measure	How / instrument	Pass criterion	REQ
V-08 (P-6)	False triggers and crosstalk	10 min idle with all 12 electrodes on objects; then touch each pad and watch neighbours	No false Note On; no crosstalk	REQ-038
V-09	Electrode readings with and without R	`SERIAL_DEBUG` deltas per object, with the 3.3 kΩ resistor and with it bridged	Touch detected on every object; delta ratio ≥ 0.904 (SIMULATION prediction)	REQ-037
V-10	ESD robustness	IEC 61000-4-2 contact discharge ±4 kV on each terminal, at a test lab	No permanent damage: all pads still detect touch afterwards	REQ-023
V-11	Series resistor after ESD	Inspect and measure R1–R12 after V-10; watch for flashover	Value within tolerance; no flashover (CR-001 if it fails)	REQ-023
V-12	JP1 bridge and etch gaps	Visual inspection with a magnifier and continuity after etching and bridging	Bridge only on the chosen side; 0.6 mm gaps clear; no short to GND in any position	REQ-025, 039
V-13	Real loop period	Toggle a spare GPIO each loop; oscilloscope	No blocking > 10 ms during play	REQ-016
V-14	Recalibration duration	GPIO toggle around `recalibrate()`	Recorded per sensor; compare with 16.8–22.1 ms (CALCULATION); no stuck notes afterwards	REQ-008, 002
V-15	Pot ADC noise on the PCB	Log `analogRead` at each zone border for 10 s	Peak-to-peak noise < 41 counts	REQ-006
V-16	Blink with live pads	Touch pads while the LED blinks scale 6; press again during a blink	Notes sent during the blink; pattern restarts with the new scale	REQ-007, 016
V-17	Button bounce and long press	Oscilloscope on the button pin; press tests	Bounce < 10 ms (ASSUMPTION); long press acts after > 800 ms	REQ-007, 008
V-18	Sensor absent or lost, and LED	Power without the module; then with it; unplug the module during play; every Pico board	No MIDI from the missing sensor, its notes released, LED at 5 Hz, retry every 2 s; 2 slow blinks when found; LED lit while touched (W boards via CYW43)	REQ-009, 010
V-19	MPR121 configuration and burst reads	Read back 0x7B–0x7F after start-up; logic analyser during play	Values as configured; 2 transactions per sensor per aftertouch pass; burst data equal to per-electrode reads	REQ-004, 037
V-20	Sampler in real browsers	Chrome and Edge on Windows and macOS with the device	Pads 36–47 sound; velocity 0 acts as Note Off; learn swaps notes	REQ-034
V-21	Sampler with a busy device	Open the device in a DAW on Windows first, then the sampler	Sampler reports the device as busy	REQ-034
V-22	Shared notes	Set an octave so two pads share a note; touch both, release one, then the other	One Note On; Note Off only after the last release; aftertouch = the stronger pad	REQ-042

11 Reproducibility

11.1 Toolchain

From [`management/TOOLCHAIN.md`] (MEASUREMENT by command, 2026-10-30) and the task handoffs:

- Host: macOS 26.5.2, arm64, Apple M4.
- KiCad 10.0.4 (kicad-cli and pcbnew Python 3.9.13).
- arduino-cli 1.5.1; arduino-pico core rp2040 6.3.1; AVR core 1.8.8; Adafruit MPR121 1.2.1, BusIO 1.17.4, MIDI Library 5.0.2, MIDIUSB 1.0.5.

- Python 3.9.6 with pytest 8.4.2 (project `.venv`); node v26.4.0; playwright-core 1.64.1 and Google Chrome 156.0.8078.74 (T-020).
- TinyTeX-1 v2026.10, pdfTeX 1.40.29 (TeX Live 2026), with siunitx, enumitem and caption added (documentation handoff T-006-DOC/T-031).
- MCP adapter `eng-tools` (`scripts/mcp/server.mjs`, @modelcontextprotocol/sdk 1.32.2).

11.2 How to rebuild

- Hardware: `sh hardware/scripts/build_all.sh` regenerates the schematic and PCB, runs ERC and DRC with schematic parity (fails on violations) and exports `hardware/fabrication/`.
- Firmware build matrix: eng-tools `firmware_compile`, or `arduino-cli compile` with the FQBN `rp2040:rp2040:<board>:usbstack=tinyusb` for rpipico, rpipicow, rpipico2, rpipico2w, each without and with `--build-property compiler.cpp.extra_flags=-DFRUTAMIDI_PCB`, plus `arduino:avr:leonardo` [`management/INTERFACES.json`].
- Host tests: eng-tools `run_pytest` with `tests/firmware` and `tests/host`.
- Systems checks: `sim/systems/budgets.py`, `interface_check.py`, `pcb_rules_check.py`; electronics calculations `sh sim/electronics/calc/run_all.sh`.
- Report: `sh reports/report/build.sh` (regenerates the requirements table from the CSV, then three pdflatex passes), or eng-tools `latex_build` with `reports/report/frutamidi_report.tex`.

TO BE FILLED FROM EVIDENCE. Released file checksums (`releases/V1_BASELINE/`), when the release exists.

A Sources

Source registers: [`management/SOURCES.md`] (S-0xx), [`requirements/SOURCES_SYSTEMS.md`] (S-xx) and [`reports/electronics/SOURCES_EE.md`] (EE-Sxx, marked "Read" or "Search"). The registers hold the full entries, access notes and checksums. Sources cited in this report:

ID	Source
S-01	Raspberry Pi Pico datasheet RP-008307-DS-2 (local copy, sha256 in the register).
S-02	Raspberry Pi documentation, "Pico-series microcontrollers", pin functions. `https://www.raspberrypi.com/documentation/microcontrollers/pico-series.html`
S-04, S-006, EE-S01	NXP/Freescale MPR121 datasheet, Rev. 4, 02/2013. `https://www.nxp.com/docs/en/data-sheet/MPR121.pdf`
S-06	Adafruit Eagle board files for the MPR121 breakout (STEMMA QT revision). `https://github.com/adafruit/Adafruit-MPR121-PCB`
S-11	Adafruit TinyUSB Arduino (in arduino-pico 6.3.1), device and MIDI sources.
S-14	USB Device Class Definition for MIDI Devices, Release 1.0 (1999-11-01). `https://www.usb.org/sites/default/files/midi10.pdf`
S-16	Raspberry Pi Pico 2 datasheet RP-008299-DS-3 (local copy).
S-17	Raspberry Pi RP2040 datasheet RP-008371-DS-1 (local copy).
EE-S08	Raspberry Pi Pico pinout. `https://pico.pinout.xyz/`
EE-S14b	Yageo leaded resistors datasheet (MFR series).
EE-S17	Vishay Beyschlag MBA/MBB/MBE professional thin-film resistors, doc 28766. `https://www.vishay.com/docs/28766/mbxsma.pdf`